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Expt.No: 6

Title: Numerical Processing using Numpy

In [1]:

```python
import numpy as np

array = np.array([1, 2, 3, 4, 5])
print("Origanal Array:")
print(array)

array_add = array + 5
array_sub = array - 2
array_mul = array * 3
array_div = array / 2
array_sqr = array ** 2

print("\nArthmetic Operations:")
print("Add 5: ", array_add)
print("Subtact 2:", array_sub)
print("Multiply by 3:", array_mul)
print("Divide by 2:", array_div)
print("Square:", array_sqr)

array_sqrt = np.sqrt(array)
array_exp = np.exp(array)

print("\nUniversal Functions:")
print("Square Root:", array_sqrt)
print("Exponential:", array_exp)

mean = np.mean(array)
std_dev = np.std(array)
sum_array = np.sum(array)

print("\nStatistical Operations:")
print("Mean:", mean)
print("Standard Deviation:", std_dev)
print("Sum:", sum_array)
```

```
Origanal Array:
[1 2 3 4 5]

Arthmetic Operations:
Add 5:  [ 6  7  8  9 10]
Subtact 2: [-1  0  1  2  3]
Multiply by 3: [ 3  6  9 12 15]
Divide by 2: [0.5 1.  1.5 2.  2.5]
Square: [ 1  4  9 16 25]

Universal Functions:
Square Root: [1.         1.41421356 1.73205081 2.         2.23606798]
Exponential: [  2.71828183   7.3890561   20.08553692  54.59815003 148.4131591 ]

Statistical Operations:
Mean: 3.0
Standard Deviation: 1.4142135623730951
Sum: 15
```

In [ ]:

```python

```